

The good news from Tata

Tata Group, one of the biggest corporate houses of India, has already announced its interest in entering the semiconductor space. While *Electronics For You* was the first media organisation to report about the appointment of Raja Manickam as the CEO of Tata's OSAT venture, Manickam during the August edition of Tech World Congress confirmed how happy he is to be associated with Tata for bringing the semiconductor ecosystem to India.

"This has been close to my heart. I have been working, living, and breathing semiconductors for the last 40 years," notes Manickam. Manickam had started a venture by the name of Tessolve Semiconductors in India in 2005. This startup was acquired by Hero Electronix in the year 2016.

Manickam reflected back towards 1995 and explained how he was looking to start semiconductor manufacturing in the country during that time. "The time around 1995 was where I thought there were many opportunities in semiconductor manufacturing and India had already missed the boat. I think the political atmosphere was not right at that time as there were so many adverse situations," said Manickam.

He added, "I am glad that it is happening now, and I am proud to be a part of it. Tata is a patient corporate house and a visionary industrialist group. I believe that the Tata OSAT is just the beginning of establishing a semiconductor ecosystem in the country."

The initiative of the Tata OSAT, as per Manickam, has been taken after serious considerations. Despite the fact that establishing an ATMP/OSAT unit does not provide return on investments quickly, Tata Group has gotten into the game considering the long-term and sustainable vectors of the business.

Another good news is that Tata Group is not the only entity investing in establishing an ATMP unit. Dr Harshad Mehta, who heads Silicon Power, is also looking to start an ATMP unit in the country under the banner of Visicon. "Our interest lies in high-power semiconductors. With Visicon the idea is to establish a silicon carbide ecosystem including silicon carbide EPI, silicon carbide processing, and

The ATMP picture in India

1. Tata Group has confirmed its foray into the OSAT space
2. The CEO of MosChip is supporting the foundation of an ATMP in Hyderabad
3. An ex-HRD minister is said to be in talks for establishing an ATMP
4. Silicon Power has started work on an ATMP facility in Gujarat

The opportunity

1. Superpowers of the world have announced investment of more than \$200 billion in semiconductors
2. The new semiconductors will need new ATMPs for testing and packaging
3. Geopolitical conditions have made corporates think about finding safer options

India - The plus points

1. Probably the largest talent pool of technical people known for their design skills
2. Almost every big semiconductor company has R&D office in India
3. Connectivity to all parts of the world
4. A government that is keen on discussions and decisions

The ATMP advantage

1. ATMPs require lower level of investment than fabs
2. ATMPs create more jobs than fabs
3. Skillset to operate machines and equipment at ATMPs can be taught at the basic level
4. ATMPs look at diploma holders in the workforce, creating more jobs at the bottom of the pyramid
5. Early commitment by potential customers

packaging," explains Dr Mehta.

Dr Mehta has already started working on Visicon's first facility located in Halol, Gujarat. The facility has bought some equipment and is waiting for more to commence operations. Dr Mehta is hopeful that the operations at the Halol facility will start before the end of this year.

"I really believe in 'why not?' and a lot of people might not know that since 2007 we have already been manufacturing power semiconductors in Baroda at a company called Ruttonsha. At Ruttonsha, we focus on power semiconductors," noted Dr Mehta. It is worth mentioning here that power semiconductors manufactured at the RattanSha facility have been used by the US Army for their power applications.

The third good news is the foundation of ASIP Tech. Venkata Simhadri, who is one of the founders of the ASIP, is also the CEO of a public listed company called MosChip. MosChip was the first fabless semiconductor company established in India.

"I would not say that setting up OSATs/ATMPs is the best way to bring the semiconductor ecosystem

to India. However, keeping the current situations prevailing in the country in mind, it is a good way. The same will also help in creating a fabless ecosystem in the country," says Venkata. ASIP will be located in Hyderabad.

Krishna Moorthy KM, CEO, India Electronics and Semiconductor Association (IESA), recalling a conference hosted by two senior vice presidents (SVPs) of National Semiconductor notes how these two SVPs had encouraged India to start looking at ATMPs first. "India should start looking first at assembly and tests, they said," recalls Krishna.

The setting up of ATMPs/OSATs is looked upon as a big building block for establishing a robust semiconductor ecosystem. There have been reports pointing towards the consumption of semiconductors touching the magical \$500 billion figure, and those figures make test and assembly (which ATMPs and OSATs do) a critical part of developing semiconductors.

The investment required for setting up an ATMP/OSAT is much lesser than what is required to establish a fabrication plant. "It's very simple economics. The return on investment